ARYAN GUPTA

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EDUCATION

Imperial College London Sep 2026 - Sep 2027
MSc Applied Computational Science & Engineering - In Progress

University of Michigan Sep 2019 - May 2023
BSE Engineering Physics, cum laude | Minors: Electrical Engineering, Computer Science, Entrepreneurship

PROFESSIONAL EXPERIENCE

Technip Energies - Houston, TX Aug 2023 - Jul 2026

Technology Developer I, Applied Physics / Carbon-Capture R&D *(Aug 2023-Jul 2025; Feb-Mar 2026)* - Built the team's first-principles MD/DFT capability across HPC and cloud; linked atomistic outputs into diffusion/CFD models and experimental workflows.

- Reduced simulation wall time by roughly two-thirds through GPU/cloud and parallel-I/O workflows; automated parameter sweeps and terabyte-scale image analysis, cutting multi-day manual analysis to hours.
- Developed uncertainty-aware particle analysis, surrogate models, active-learning/optimization workflows, and verification tests; authored 15 reviewed technical notes and a 70-page market/business analysis used in funding and roadmap decisions.

Americas AI Solutions Initiative - Co-Founder, Co-Lead & Technical Lead *(Dec 2025-Jul 2026)* - Co-led an internal initiative spanning 4 operating centers and 8 engineering functions while remaining in an engineering role.

- Scoped and led engineering AI workflows across structured data and technical documents, including natural-language-to-SQL and vendor-document analysis; one bid-review workflow reduced manual review effort by about 80%.

Specialist Engineer I - Instrumentation & Controls *(Jul 2025-Jul 2026)* - Worked on hydrogen, carbon-capture, and brownfield scopes; reviewed instrumentation/control documentation against P&IDs, PFDs, Cause & Effect diagrams, narratives, and standards.

SELECTED RESEARCH & OPEN SOFTWARE

WaveSleuth-Devito: Inverse Wave Playground v1.0 Jun-Aug 2026
Devito-backed 2D acoustic inverse-problem sandbox | GitHub: guptaaryanr/wavesleuth-devito

- Built deterministic forward simulation, sparse acquisition, circle/ellipse/coarse-mask inversion baselines, blind challenges, uncertainty summaries, active sensing, scoring, and release validation.
- Canonical v1.0 toy challenges reached IoU 1.000 (mask), 0.895 (circle radius+velocity), 0.800 (circle), and 0.787 (ellipse); deliberately scoped as an inspectable sandbox rather than a field-scale inversion claim.

Trajectory-Only Structural Regime Detection Feb-May 2026
Preprint DOI: 10.5281/zenodo.20278240 | Code DOI: 10.5281/zenodo.19465797

- Developed a trajectory-only detector for structural changes in deterministic nonlinear systems, keeping model-aware transition definitions separate for post-hoc evaluation.
- Tested on FitzHugh-Nagumo and forced van der Pol systems with boundary extraction, lead-distance, nuisance, specificity, robustness, and ablation studies.

ThermoBench-Consist v1.0 Sep-Oct 2025
Preprint DOI: 10.5281/zenodo.17489426 | Code DOI: 10.5281/zenodo.17330441

- Built a CPU-only diagnostic for ML thermodynamic surrogates using density-pressure monotonicity, positive compressibility, Clapeyron consistency, and speed-of-sound checks, with guardrails for fragile thermodynamic regions.

Assessing the Limits of GNNs for Vapor-Liquid Equilibrium May-Sep 2025

arXiv:2509.10565

Negative-results study showing that acceptable pointwise prediction errors can still produce derivative pathologies and liquid-phase inconsistencies that make learned surrogates unusable inside VLE solvers.

Energy-Efficient Robotics Software: Systematic Literature Review Jan-Aug 2025

arXiv:2508.12170 | Replication DOI: 10.5281/zenodo.16907564

Synthesized 79 post-2020 studies, proposed a taxonomy of energy-aware techniques and a reporting checklist, and released the screening/figure-generation replication package.

REB-1: Robot Energy Benchmark v0.1 Aug 2025

Code DOI: 10.5281/zenodo.17204853

Built a lightweight CLI/notebook benchmark for reproducible power traces, Wh integration, and user-specified grid-intensity conversion to gCO2 for quick autonomy-workload A/B tests.

EARLIER RESEARCH, TEACHING & BUILDING

BIRDS Lab, University of Michigan - Research Assistant *(2021-2023)* - Built and debugged adaptable quadcopter hardware/software across ESP32/FTDI telemetry, power electronics, flight-control software, and mechanical integration.

EECS 314 Circuit Design & Analysis - Instructional Aide *(2022-2023)* - Led discussion groups for about 80 students and a 20-student circuits lab; supported course delivery for 160+ students.

MiTEE - Plasma Team / Structures Liaison *(2020)* - Worked on electrodynamic-tether propulsion feasibility for pico/nanosatellites, including thermodynamic analysis, electronics prototyping, and subsystem interfaces.

HelixCases - Co-Founder & Software Developer *(2022)* - Co-built a university technical-recruiting competition platform reaching 550+ students at 20 universities; secured a $10k Lyft contract.

SELECTED ENGINEERING PROJECTS

Electric Longboard Remote Controller *(2022)* - STM32L010C6T6 remote with UART/LPUART phone link, battery/display feedback, modified VESC firmware, and a 3D-printed enclosure.

Out-of-Order Processor Architecture *(2022)* - MIPS R10K-style SystemVerilog core with 2-way superscalar issue, I/D caches, LSQ, branch prediction, precise exceptions, baseline SMT, and a visual debugger.

Low-Autonomy Follower Robot *(2022)* - Two-Zumo leader/follower system using Nucleo boards, LED-encoded commands, PixyCam vision, ultrasonic ranging, and PID control.

LEADERSHIP & SERVICE

Project RISHI - Fundraising Co-Chair *(2020-2021)* - Helped secure $16k toward a $35k budget through grant applications and campus fundraising.

StartUM - Member & Mentor *(2021-2022)* - Mentored an early-stage medical-volunteer startup through validation and a change in target market.

Citizens for Animal Protection - Volunteer *(2023-2024)* - Dog walking, kennel cleaning, and day-to-day shelter support in Houston.

SKILLS

Scientific computing & simulation: Python, C/C++, MATLAB/Simulink, NumPy, SciPy, LAMMPS, NWChem, Devito, MD/DFT, ODE/PDE simulation, thermodynamics, UQ

ML, optimization & evaluation: PyTorch, scikit-learn, OpenCV, surrogate modeling, active learning, numerical optimization, benchmarking, validation/evaluation harnesses

Systems & tooling: Linux/Unix, Git, Docker, CI/GitHub Actions, SLURM, GCP, SQL/PostgreSQL, OCR/PDF parsing, embedded STM32/ARM, ESP32, UART, FreeRTOS, PID control

HONORS & CREDENTIALS

Six Sigma Green Belt (CSSGB), Technip Energies (2026) | Foundational Finance, University of Michigan (2020) | Cum Laude, University of Michigan (2023) | University Honors & Dean's List (2019, 2020)